Begin

Nicholas Alan Gilbert Emperical Exercise #2 10-26-26

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Set Up

We will import necessary libraries and custom helper-functions written in another file.

In [1]:
```
%load_ext autoreload
%autoreload 2

import yfinance as yf
import pandas as pd
import matplotlib.pyplot as plt
import scipy.stats as st
import numpy as np
import ipywidgets

#Custom Helper Functions
import helper.tools as tools
```

In [3]:
```
stocks = tools.StockSearch()
```

```
VBox(children=(HBox(children=(Combobox(value='', description='Ticker:',
options=('MMM', 'AOS', 'ABT', 'ABBV', …
```

In [13]:
```
stocks.start
```

Out[13]:
```
datetime.date(2000, 1, 1)
```

In [14]:
```
stocks.end
```

Out[14]:
```
datetime.date(2026, 10, 26)
```

In [18]:
```
stocks.tickers = ['XLC', 'XLY', 'XLP', 'XLE', 'XLF', 'XLV', 'XLI', 'XL

stocks.tickers
```

Out[18]:
```
['XLC', 'XLY', 'XLP', 'XLE', 'XLF', 'XLV', 'XLI', 'XLB', 'XLRE', 'XL
K', 'XLU']
```

In [19]:
```
df = stocks.data.copy()
```

```
[**********100%**********]  11 of 11 completed
```

In [20]:
```
df
```

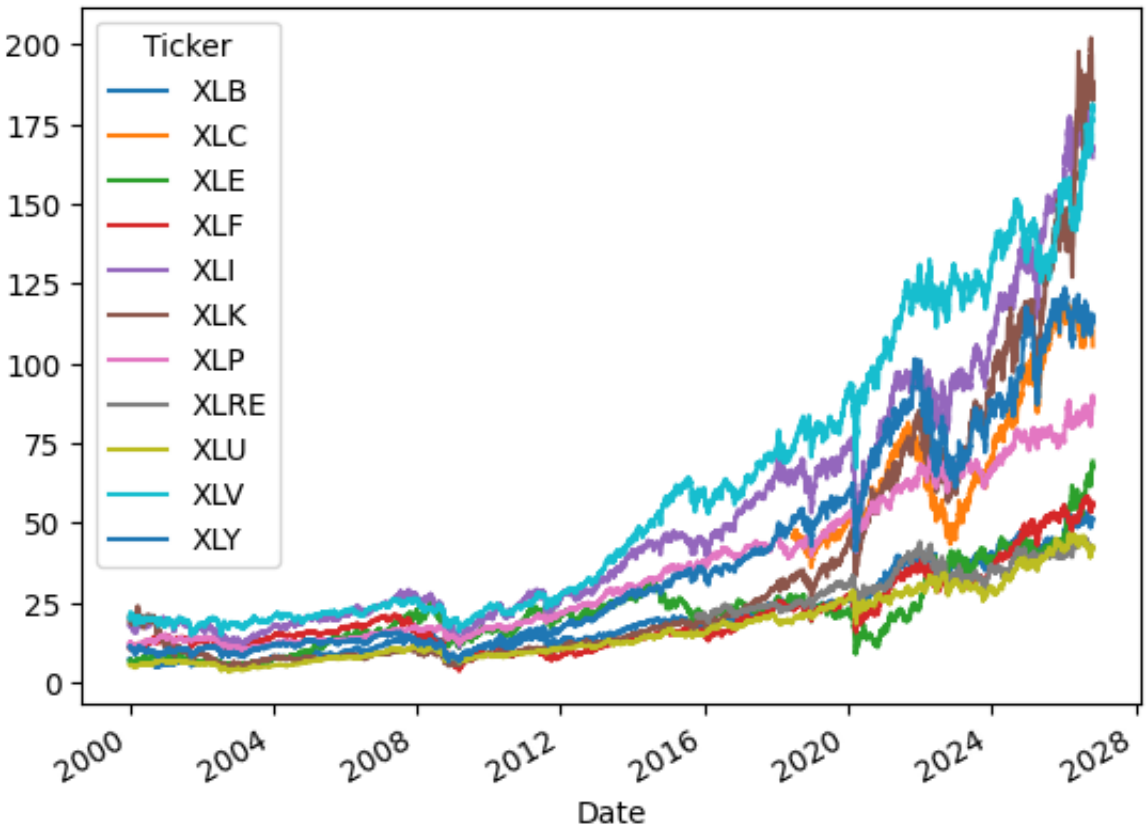
Out[20]:

Price						
Ticker	**XLB**	**XLC**	**XLE**	**XLF**	**XLI**	**XLK**
Date						
2000-01-03	7.168345	NaN	6.569214	11.013208	17.982185	20.577808
2000-01-04	7.099332	NaN	6.445558	10.531755	17.485386	19.533834
2000-01-05	7.392622	NaN	6.615584	10.449008	17.407461	19.243847
2000-01-06	7.582393	NaN	6.870625	10.907898	17.641241	18.605864
2000-01-07	7.578085	NaN	6.944041	11.088442	18.313396	18.930651
...	...	...	...	...	...	...
2026-10-19	50.950001	110.089996	67.800003	55.060001	166.679993	185.960007
2026-10-20	51.230000	110.190002	67.970001	55.720001	166.800003	185.440002
2026-10-21	51.320000	108.970001	68.160004	56.040001	168.309998	188.539993
2026-10-22	51.689999	107.419998	67.570000	56.650002	166.729996	182.339996
2026-10-23	51.310001	105.510002	69.330002	55.900002	164.619995	183.229996

6743 rows × 22 columns

In [21]:
```
df['Close'].plot()
```

Out[21]: <Axes: xlabel='Date'>



Data Handling

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Log Returns

Log returns are computed under the assumption of *continous time*. It comes from the inverse of the exponential function e^t

$\ln(e^t) = t$

Read: the natural-log of e *to the something* is that something.

Context

Where We Definied Returns Before;

$ r_t = $ return at time period t

$ p_t = $ asset price at time t

$ t = $ time-period eg. daily, weekly, monthly

Asset Return:

$$ r_t = \frac{p_t}{p_{t-1}} - 1$$

Then a series of successive returns of n periods compounded is represented geometrically:

$$r_{\text{cumulative}} := (\prod_{i=1}^{n} 1 + r_i) - 1 = [(1+r_1) \cdot (1+r_2) \cdot (1+r_3) \dots (1+r_n)] - 1$$

This is a more expensive operation, adding 1s to each entry then performing n multiplications.

Natural Log to the Rescue

Some nice arithmetic properties of using ln() returns is that compounding is additive.

$$\ln\left(\frac{p_t}{p_{t-1}} \right) = \ln(p_t) - \ln(p_{t-1})$$

$$\ln\left({p_t}\cdot{p_{t-1}} \right) = \ln(p_t) + \ln(p_{t-1})$$

We Define Log-Returns:

$$r_t := \ln(p_t) - \ln(p_{t-1})$$

$$r_t := \ln(p_{\text{today} }) - \ln(p_{\text{yesterday} })$$

Calculation Method

We can apply NumPy's `ln()` function to an entire array of asset prices, then use pandas' `.shift()` method to subract a whole shifted array from the other.

Will look something like this:

```
log_returns = np.log(data['Close']) -
np.log(data['Close']).shift(1)
```

Missing Data

At this point we will 'produce' a NaN value in our return calculation from our first date, we don't have a previous data point to compute a return from that point. So we'll just drop it. Other stock's will exhbit NaN's, say they weren't listed on an exchange at that point, so no data occurs for them.

Interpretation

Working with log-returns is nice mathematically. But it will differ from the true return an investor will see if they have their money invested. For example, between 2018-02-26 and 2026-08-28.

Example

AAPL's Average Daily Log-Return: 0.096%

AAPL's Average Daily Real Return: 0.123%

In [ ]:

Task 1

Compute Log-Returns

In [22]:
```
log_returns = np.log(df['Close']) - np.log(df['Close']).shift(1)
log_returns
```

Out[22]:

Ticker	XLB	XLC	XLE	XLF	XLI	XLK	
Date							
2000-01-03	NaN	NaN	NaN	NaN	NaN	NaN	
2000-01-04	-0.009674	NaN	-0.019003	-0.044700	-0.028016	-0.052065	-0.0
2000-01-05	0.040482	NaN	0.026037	-0.007888	-0.004467	-0.014957	0.
2000-01-06	0.025346	NaN	0.037827	0.042980	0.013340	-0.033715	0.
2000-01-07	-0.000568	NaN	0.010629	0.016416	0.037393	0.017306	0.
...	...	...	...	...	...	...	
2026-10-19	0.000589	-0.002087	-0.006469	-0.008320	-0.006459	0.013480	-0.0
2026-10-20	0.005481	0.000908	0.002504	0.011916	0.000720	-0.002800	0.
2026-10-21	0.001755	-0.011134	0.002791	0.005727	0.009012	0.016579	-0.0
2026-10-22	0.007184	-0.014326	-0.008694	0.010826	-0.009432	-0.033437	0.0
2026-10-23	-0.007379	-0.017941	0.025714	-0.013328	-0.012736	0.004869	-0.

6743 rows × 11 columns

In [23]:
```
log_returns.isna().sum(), log_returns[log_returns.index > '2018'].isna
```

Out[23]:
```
(Ticker
 XLB        1
 XLC     4645
 XLE        1
 XLF        1
 XLI        1
 XLK        1
 XLP        1
 XLRE    3967
 XLU        1
 XLV        1
 XLY        1
 dtype: int64,
 Ticker
 XLB       0
 XLC     117
 XLE       0
 XLF       0
 XLI       0
 XLK       0
 XLP       0
 XLRE      0
 XLU       0
 XLV       0
 XLY       0
 dtype: int64)
```

Note on NaN Handling

So far we can just ignore the NaNs. They aren't 'in the way' for calculating log-returns at this point. The 'sanity check' above tells us there isn't large holes in the data. I know that ticker TRMD wasn't listed until 2018 compared to our other tickers. So we are missing data for historical reasons. Plus we will have NaN from calculating returns.

If we know some caclulations will choose to 'ignore NaNs' or 'include NaNs', this is what looking up the code documentation is for.

In [24]:
```
sum_stats = tools.summary_statistics_table(log_returns)

sum_stats
```

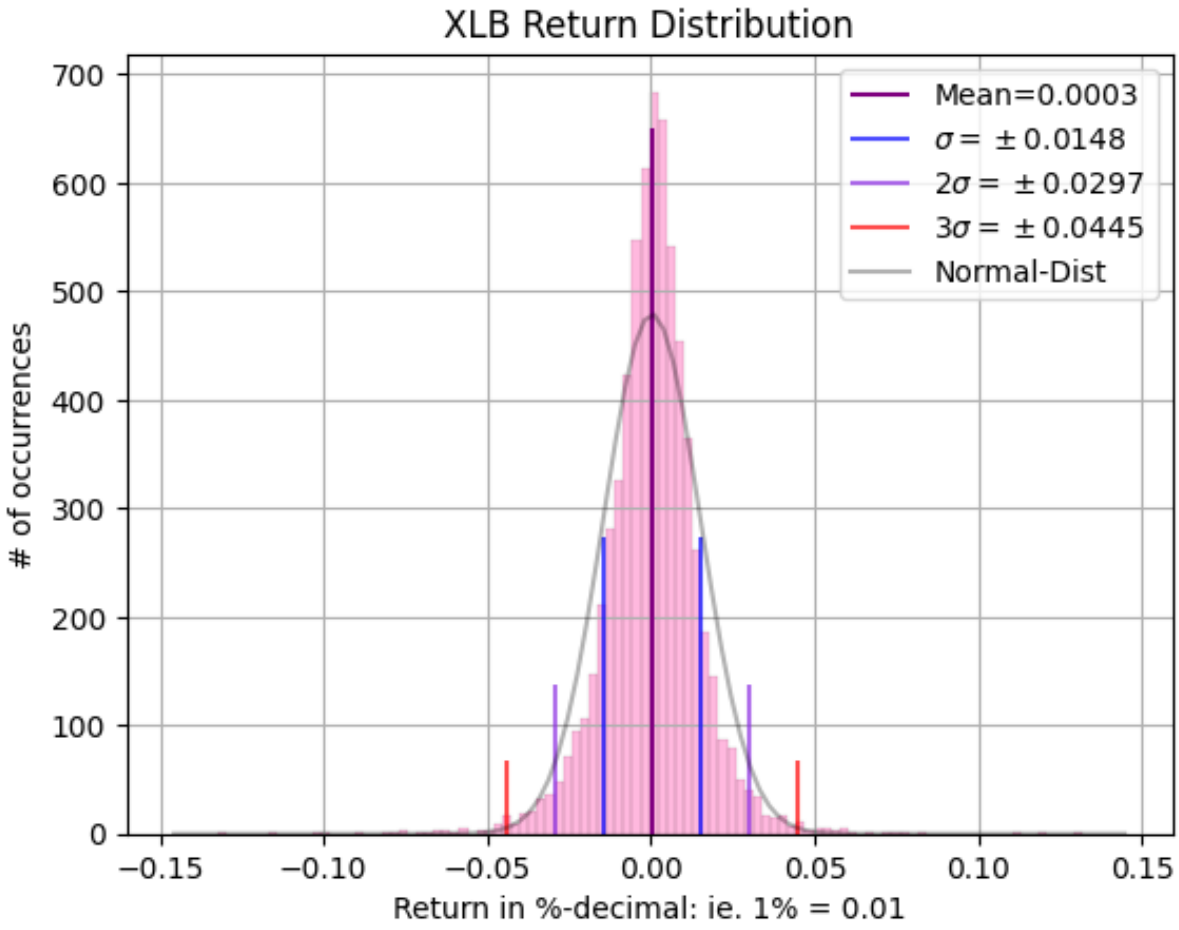
Out[24]:

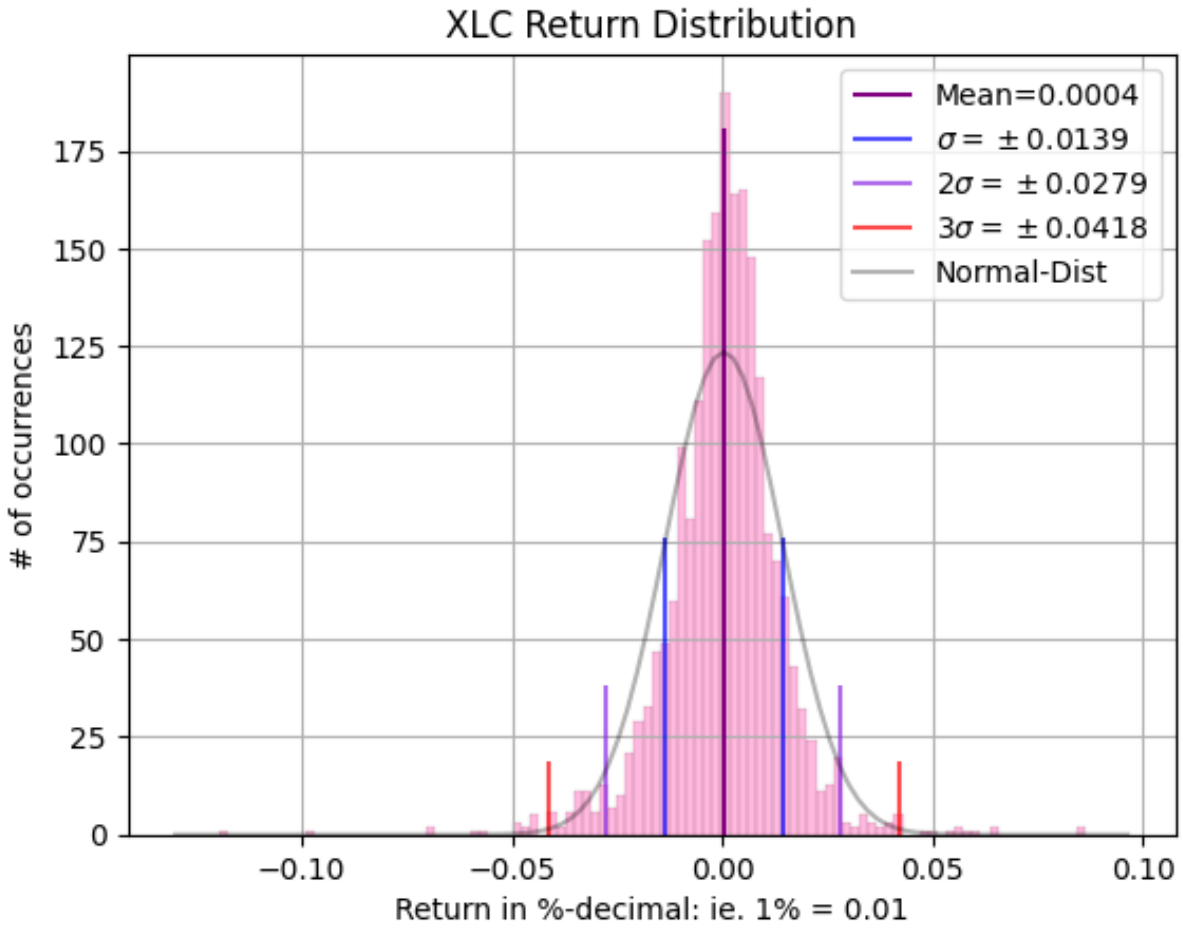
	XLB	XLC	XLE	XLF	XLI	XLK
Mean	0.000292	0.000395	0.000350	0.000241	0.000328	0.000324
Median	0.000745	0.001067	0.000821	0.000616	0.000813	0.001063
Standard-Deviation	0.014826	0.013948	0.018165	0.017757	0.013335	0.016288
Skewness	-0.211880	-0.456690	-0.649857	-0.159880	-0.332843	0.080895
Kurtosis	6.730035	7.106145	12.215804	15.469194	7.993847	7.074342
Min	-0.132527	-0.119679	-0.224910	-0.182322	-0.120408	-0.148662
Max	0.131534	0.086087	0.152504	0.152394	0.119126	0.149296

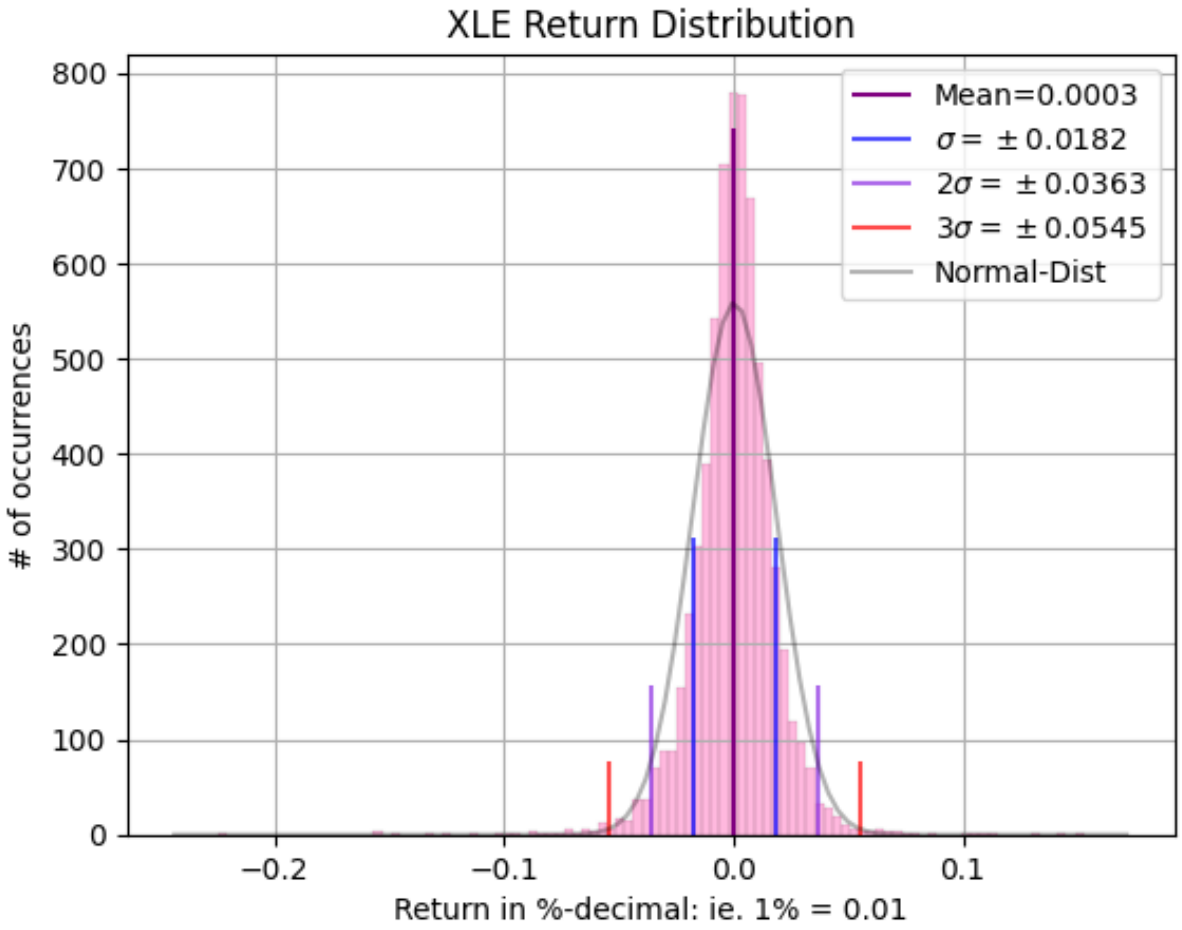
In [25]:
```
np.exp(sum_stats.loc['Mean']) - 1
```

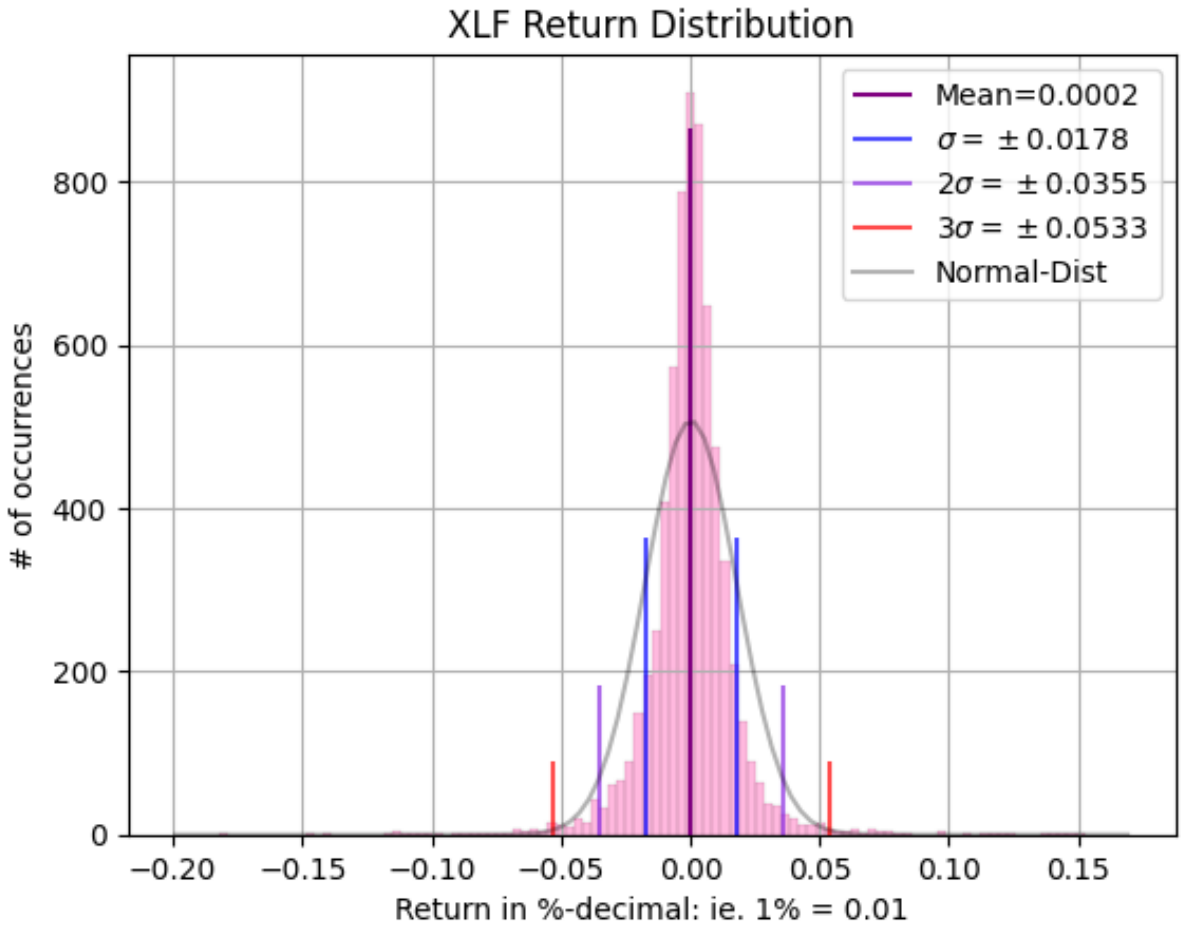
Out[25]:
```
XLB     0.000292
XLC     0.000395
XLE     0.000350
XLF     0.000241
XLI     0.000328
XLK     0.000324
XLP     0.000296
XLRE    0.000260
XLU     0.000305
XLV     0.000323
XLY     0.000344
Name: Mean, dtype: float64
```

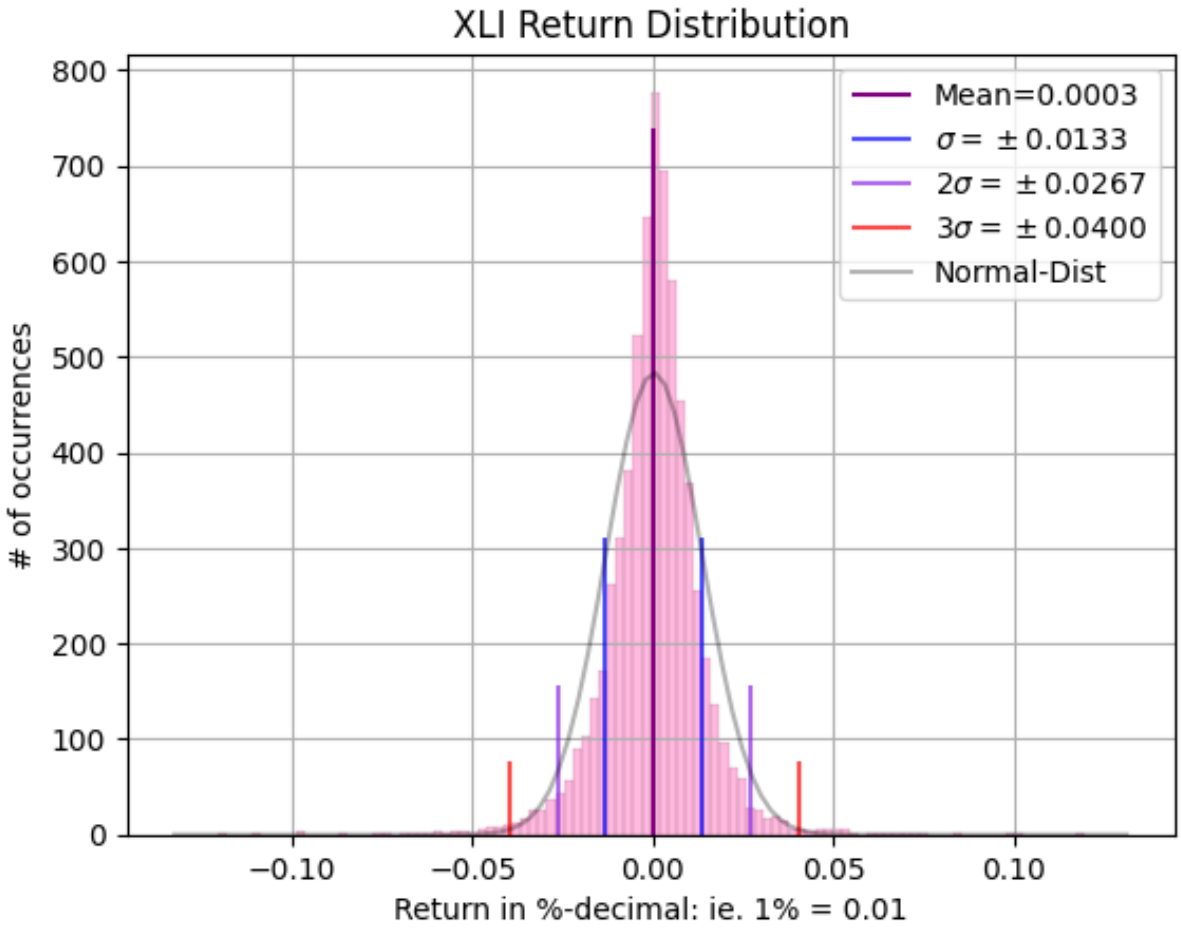
In [27]:
```
tools.return_histograms(log_returns)
```

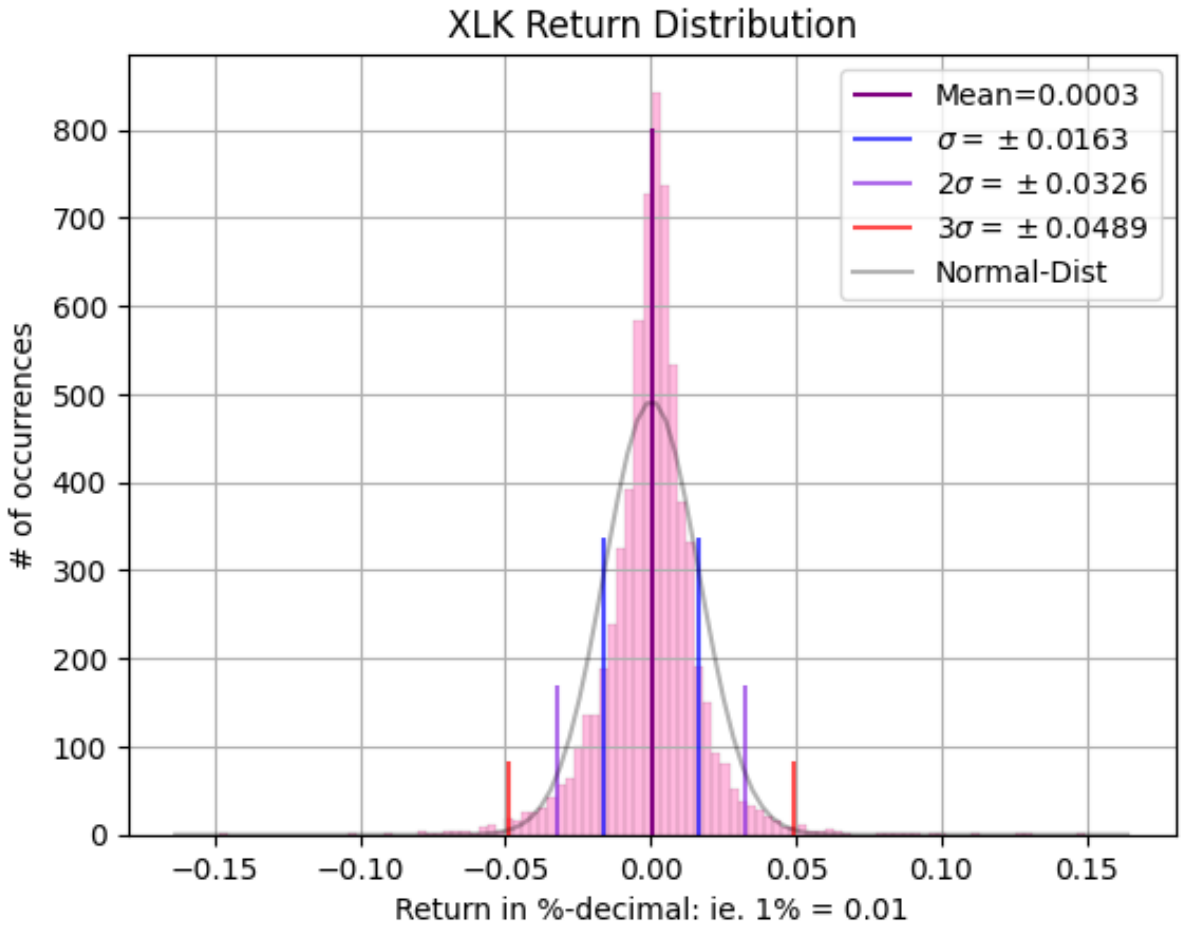

XLB Return Distribution
Mean=0.0003
σ = ±0.0148
2σ = ±0.0297
3σ = ±0.0445
Normal-Dist
of occurrences
0
100
200
300
400
500
600
700
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
Return in %-decimal: ie. 1% = 0.01

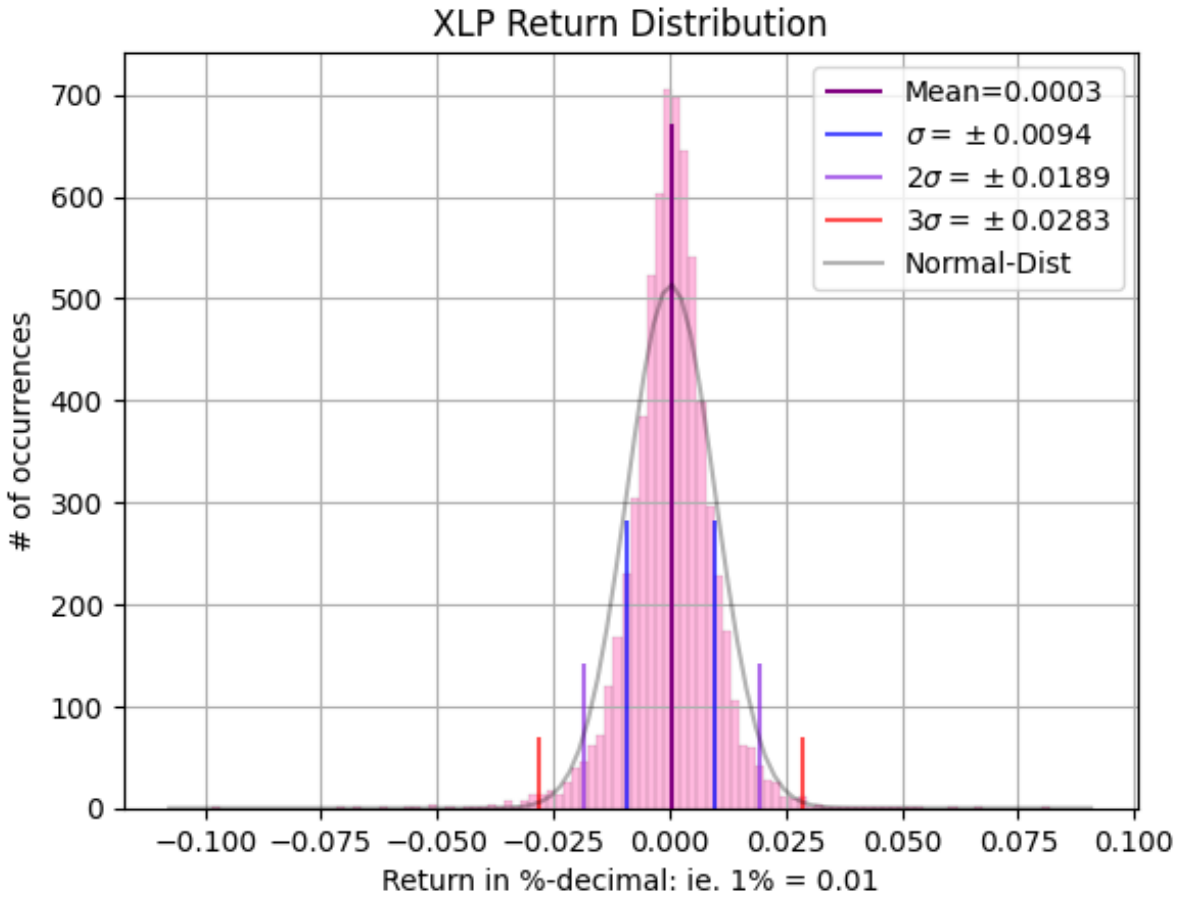

XLC Return Distribution
Mean=0.0004
σ = ±0.0139
2σ = ±0.0279
3σ = ±0.0418
Normal-Dist
of occurrences
0
25
50
75
100
125
150
175
−0.10
−0.05
0.00
0.05
0.10
Return in %-decimal: ie. 1% = 0.01

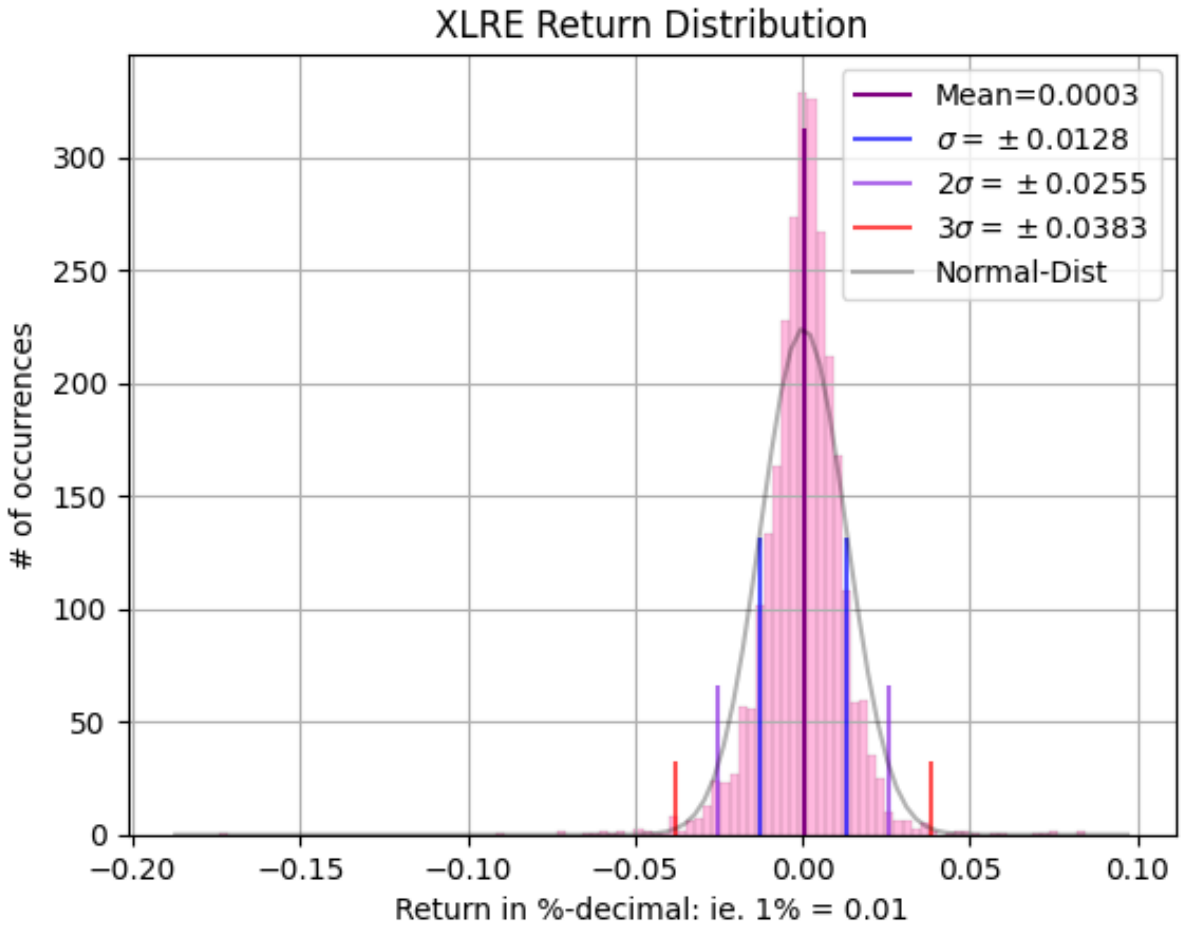

XLE Return Distribution
Mean=0.0003
$\sigma = \pm 0.0182$
$2\sigma = \pm 0.0363$
$3\sigma = \pm 0.0545$
Normal-Dist
of occurrences
800
700
600
500
400
300
200
100
0
−0.2
−0.1
0.0
0.1
Return in %-decimal: ie. 1% = 0.01

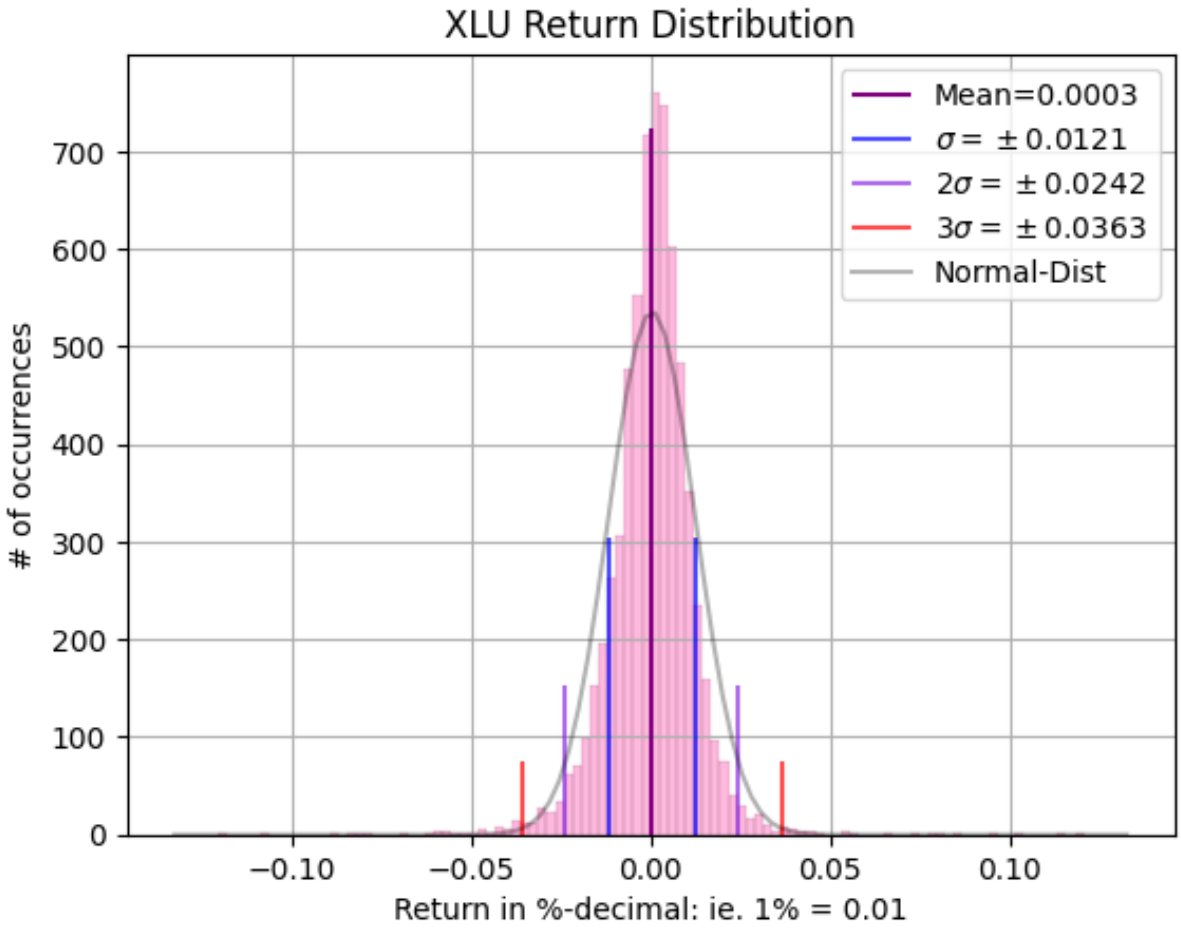

XLF Return Distribution
Mean=0.0002
$\sigma = \pm 0.0178$
$2\sigma = \pm 0.0355$
$3\sigma = \pm 0.0533$
Normal-Dist
of occurrences
0
200
400
600
800
−0.20
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
Return in %-decimal: ie. 1% = 0.01

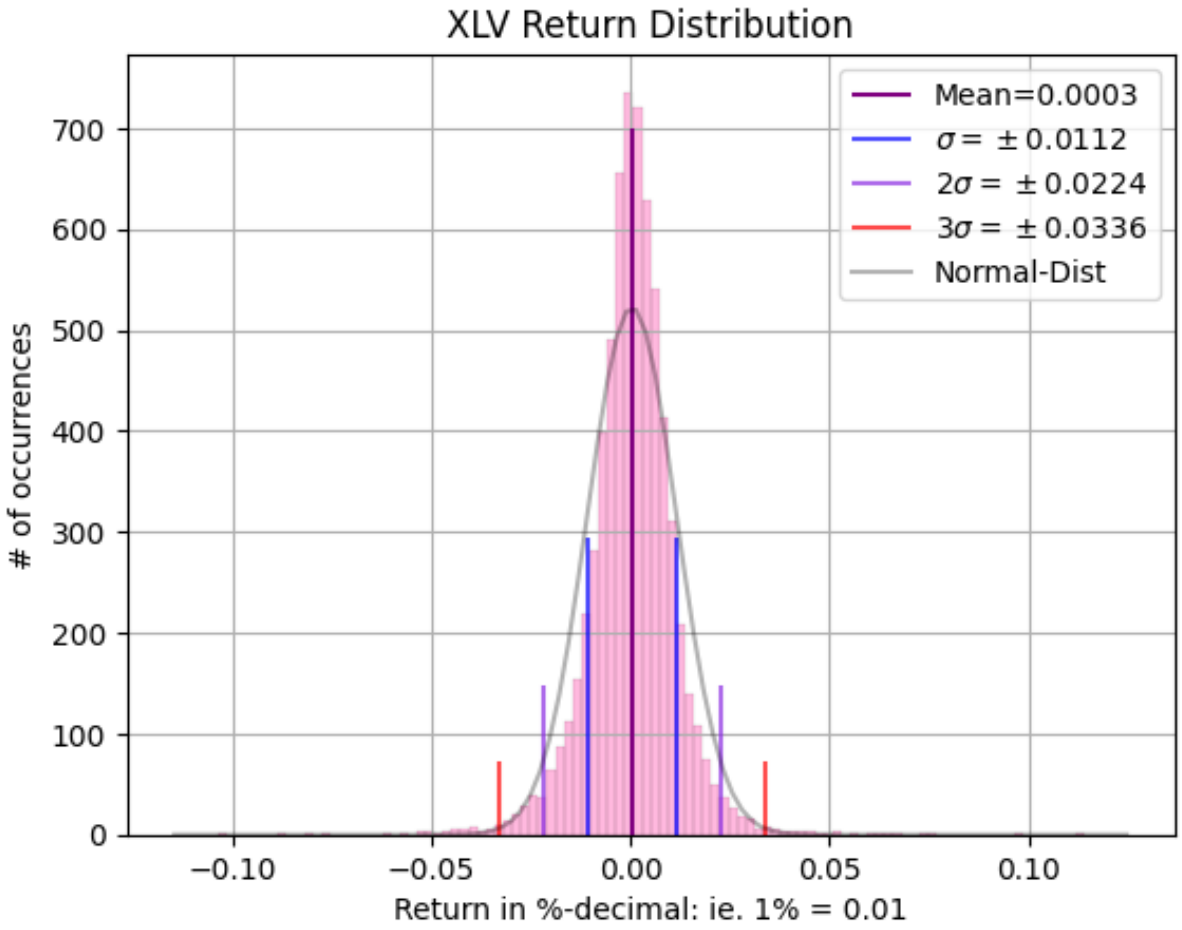

XLI Return Distribution
Mean=0.0003
$\sigma = \pm 0.0133$
$2\sigma = \pm 0.0267$
$3\sigma = \pm 0.0400$
Normal-Dist
of occurrences
0
100
200
300
400
500
600
700
800
−0.10
−0.05
0.00
0.05
0.10
Return in %-decimal: ie. 1% = 0.01

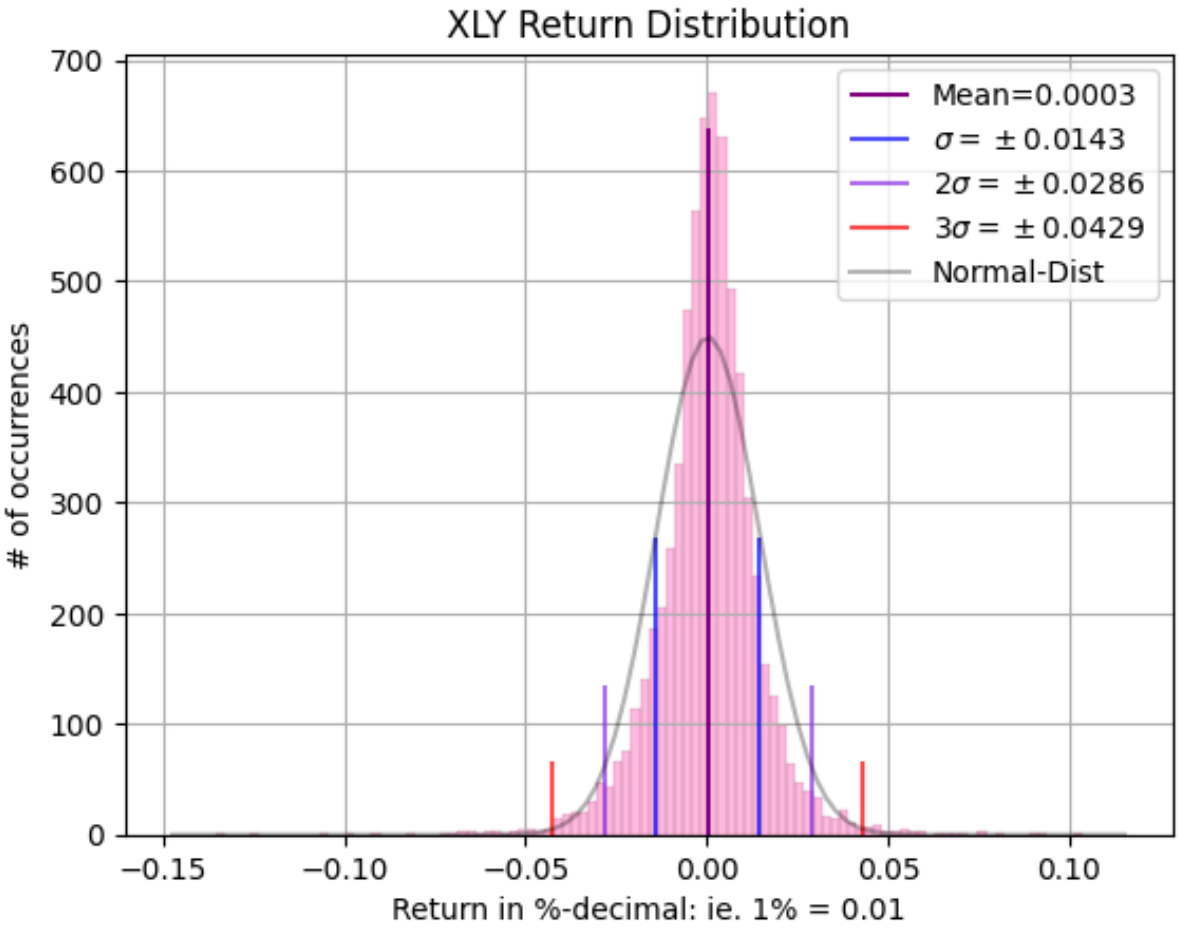

XLK Return Distribution
Mean=0.0003
σ = ±0.0163
2σ = ±0.0326
3σ = ±0.0489
Normal-Dist
of occurrences
0
100
200
300
400
500
600
700
800
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
Return in %-decimal: ie. 1% = 0.01


XLP Return Distribution
Mean=0.0003
σ = ±0.0094
2σ = ±0.0189
3σ = ±0.0283
Normal-Dist
of occurrences
0
100
200
300
400
500
600
700
−0.100
−0.075
−0.050
−0.025
0.000
0.025
0.050
0.075
0.100
Return in %-decimal: ie. 1% = 0.01


XLRE Return Distribution
Mean=0.0003
$\sigma = \pm 0.0128$
$2\sigma = \pm 0.0255$
$3\sigma = \pm 0.0383$
Normal-Dist
of occurrences
0
50
100
150
200
250
300
−0.20
−0.15
−0.10
−0.05
0.00
0.05
0.10
Return in %-decimal: ie. 1% = 0.01


XLU Return Distribution
Mean=0.0003
σ = ±0.0121
2σ = ±0.0242
3σ = ±0.0363
Normal-Dist
of occurrences
0
100
200
300
400
500
600
700
−0.10
−0.05
0.00
0.05
0.10
Return in %-decimal: ie. 1% = 0.01


XLV Return Distribution
Mean=0.0003
$\sigma = \pm 0.0112$
$2\sigma = \pm 0.0224$
$3\sigma = \pm 0.0336$
Normal-Dist
of occurrences
0
100
200
300
400
500
600
700
−0.10
−0.05
0.00
0.05
0.10
Return in %-decimal: ie. 1% = 0.01



Task 2

Annual Volatility

Since log-returns are additive for compounding. We just need to group by year and sum for annualized returns across our time period.

Volatility I will elect to take rolling 90-day windows daily, then compute the average annually, then scale up to annualized by multiplying by $\sqrt{252}.$ Because volatility scales by $\sqrt{\text{time}}$.

In [28]:
```
df_vol = log_returns.groupby(log_returns.index.year).std() * np.sqrt(2
df_vol
```

Out[28]:

Ticker	XLB	XLC	XLE	XLF	XLI	XLK	XLP
Date							
2000	0.309386	NaN	0.285607	0.331983	0.245537	0.434862	0.250427
2001	0.270129	NaN	0.258497	0.252794	0.269053	0.469465	0.188387
2002	0.288892	NaN	0.288284	0.311292	0.282543	0.417713	0.194693
2003	0.187335	NaN	0.159788	0.197232	0.176811	0.251905	0.135605
2004	0.170972	NaN	0.178713	0.122632	0.127967	0.173093	0.098850
2005	0.159742	NaN	0.256584	0.118993	0.117858	0.121619	0.090383
2006	0.178026	NaN	0.249429	0.116614	0.120169	0.141538	0.087004
2007	0.225392	NaN	0.250980	0.240719	0.160724	0.163587	0.108563
2008	0.471753	NaN	0.614376	0.715477	0.384242	0.408696	0.241994
2009	0.337755	NaN	0.352526	0.642079	0.327775	0.257241	0.172573
2010	0.249590	NaN	0.234202	0.254628	0.220508	0.187157	0.121173
2011	0.310824	NaN	0.318000	0.334177	0.274064	0.226509	0.148335
2012	0.176333	NaN	0.182085	0.175048	0.153067	0.147316	0.093845
2013	0.143683	NaN	0.143047	0.143168	0.129368	0.115308	0.113382
2014	0.141166	NaN	0.184764	0.128607	0.137210	0.126536	0.096439
2015	0.184930	NaN	0.248918	0.176189	0.160834	0.177486	0.138115
2016	0.178331	NaN	0.245992	0.187039	0.146979	0.153373	0.118053
2017	0.104121	NaN	0.132872	0.130699	0.091853	0.103987	0.076377
2018	0.191717	0.223323	0.222950	0.196180	0.190987	0.233688	0.145219
2019	0.156700	0.164203	0.189900	0.155148	0.155862	0.180847	0.107287
2020	0.385618	0.334391	0.611573	0.455230	0.397672	0.403767	0.278610
2021	0.170327	0.162863	0.295374	0.188600	0.156613	0.192899	0.111013
2022	0.252493	0.309286	0.354034	0.244616	0.218981	0.327994	0.176506
2023	0.168363	0.195009	0.229192	0.162461	0.147544	0.184957	0.113827
2024	0.136177	0.154466	0.179006	0.140974	0.136568	0.222421	0.098182
2025	0.199776	0.184982	0.248423	0.191217	0.187933	0.274180	0.138412
2026	0.183233	0.171835	0.230520	0.149908	0.180457	0.277967	0.159132

In [29]:
```
df_yearly = df['Close'].groupby(df.index.year)
```

In [30]:
```
yearly_returns = {}

for year in df_yearly:
    yr = year[0]
    df1 = year[1]


    lg_rtn = np.log(df1) - np.log(df1).shift(1)

    yearly_returns.update({yr: lg_rtn.sum()})

yearly_return_summary = pd.DataFrame(yearly_returns).T
yearly_return_summary = np.exp(yearly_return_summary) - 1
```

Annual Return Summary

Below is a table of annual returns.

Notable Years:

- 2000 was the Dot-Com era crash that led to large negative returns in technology companies like Apple.
- 2008 was the Great Financial Crisis, so we see stocks with a large negative return that year
- 2020, although Intra-Year 2020 saw a massive crash in Q1, the stock market recovered and rallied as a whole throughout the year as liquidity was pumped in the system and new market participants took discretionary money on stock trading, (ie. Robinhood stock-pickers).
- 2022 we can see the 'Mini Bear Market' of interest rate hiking following COVID

Calculation:

To take log-returns back into percentage returns, we need to exponentiate them.

$$r_{\text{Compounded}} = e^{r_{\text{log}} } - 1$$

In [31]:
```
yearly_return_summary
```

Out[31]:

Ticker	XLB	XLC	XLE	XLF	XLI	XLK	
2000	-0.142583	0.000000	0.268888	0.307566	0.095412	-0.435175	0.2
2001	0.041926	0.000000	-0.183253	-0.068147	-0.063743	-0.186440	-0.0
2002	-0.060304	0.000000	-0.140377	-0.146387	-0.245730	-0.396750	-0.1
2003	0.339032	0.000000	0.232182	0.259131	0.285487	0.315640	0.0
2004	0.140480	0.000000	0.339839	0.110175	0.178394	0.051218	0.0
2005	0.051411	0.000000	0.455487	0.065884	0.036494	0.005408	0.0
2006	0.159147	0.000000	0.127144	0.169318	0.127622	0.099993	0.1
2007	0.228969	0.000000	0.418480	-0.195698	0.127845	0.151106	0.
2008	-0.435069	0.000000	-0.390705	-0.539958	-0.376934	-0.402979	-0.1
2009	0.429407	0.000000	0.159852	0.162653	0.177088	0.458361	0.1
2010	0.169050	0.000000	0.180497	0.096949	0.254285	0.097669	0.1
2011	-0.116350	0.000000	0.020764	-0.189217	-0.018037	0.014858	0.
2012	0.115954	0.000000	0.023694	0.251425	0.124586	0.136882	0.1
2013	0.229103	0.000000	0.235139	0.316712	0.372531	0.221757	0.2
2014	0.080480	0.000000	-0.073594	0.156887	0.119293	0.190079	0.
2015	-0.088247	0.000000	-0.218634	-0.017701	-0.042048	0.056737	0.0
2016	0.186546	0.000000	0.280678	0.248247	0.216346	0.165355	0.0
2017	0.232933	0.000000	-0.019934	0.206506	0.232463	0.330732	0.1
2018	-0.160699	-0.168805	-0.195426	-0.130832	-0.137602	-0.028770	-0.0
2019	0.234935	0.293248	0.095823	0.307759	0.284266	0.497592	0.
2020	0.218895	0.254309	-0.332715	-0.026848	0.088475	0.409737	0.1
2021	0.286644	0.177063	0.530376	0.366520	0.240806	0.369775	0.1
2022	-0.111397	-0.380601	0.593854	-0.116722	-0.047858	-0.284057	-0.0
2023	0.125718	0.508717	0.029843	0.116392	0.178772	0.575120	-0.0
2024	0.003252	0.354711	0.044381	0.300030	0.185053	0.249011	0.1
2025	0.111721	0.224838	0.066315	0.151884	0.197651	0.249014	0.0
2026	0.126820	-0.089206	0.548828	0.030106	0.050345	0.274365	0.

Task 3

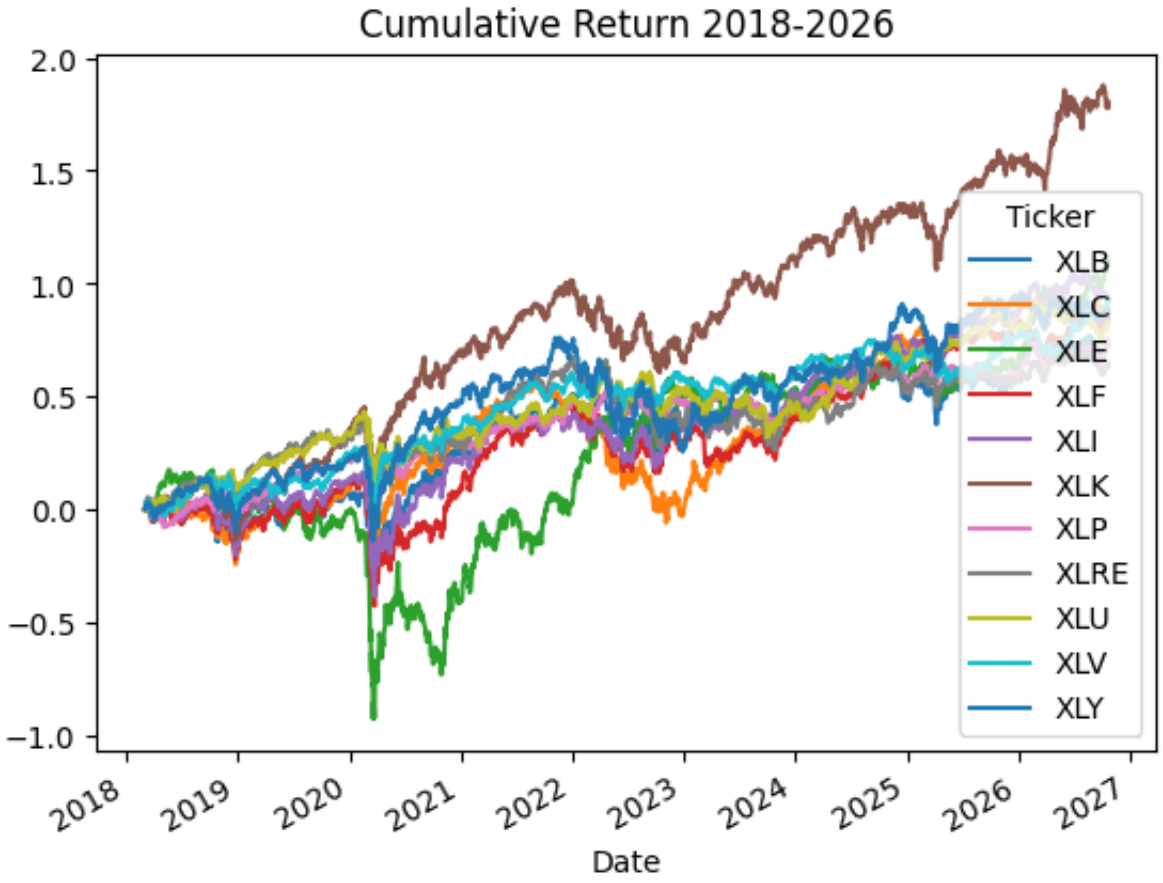
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Data Visualization

Data Visualization

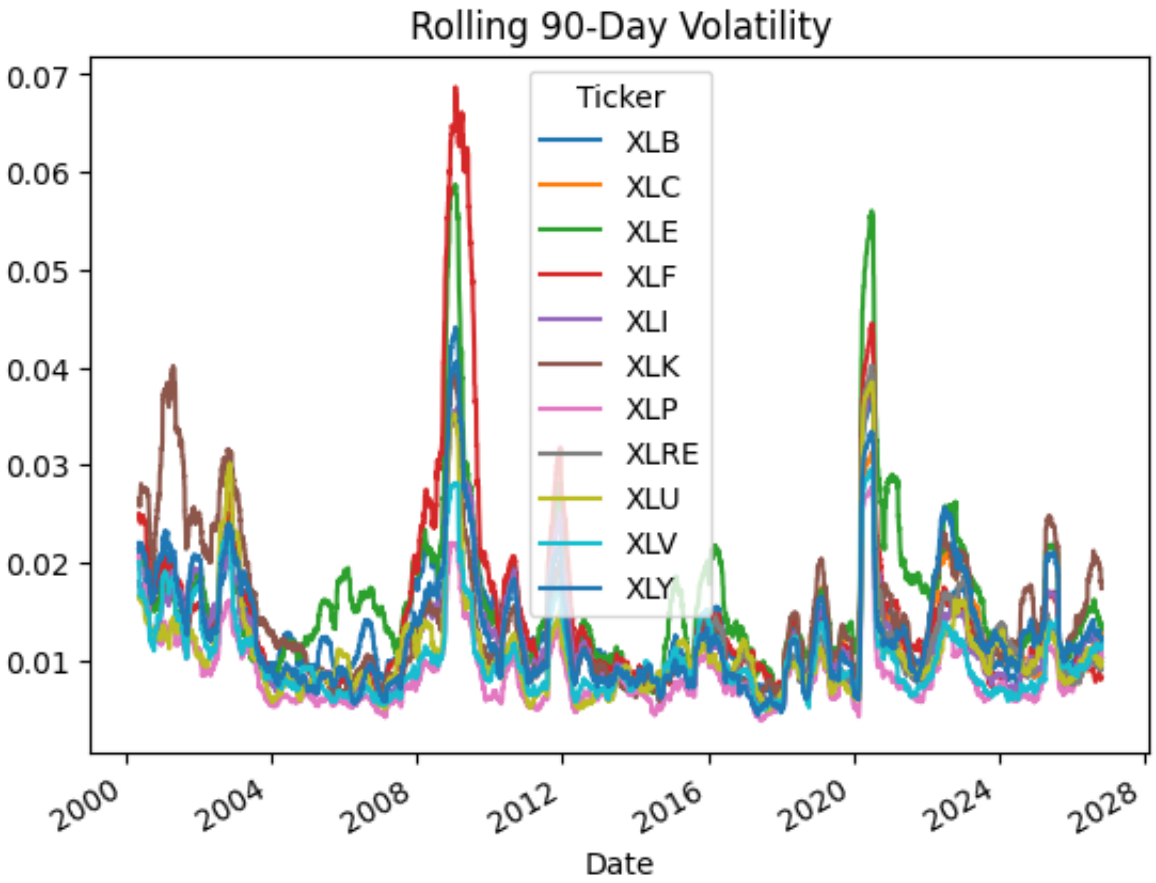
In [32]:
```
log_returns[log_returns.index > '03-01-2018'].cumsum().plot(title='Cum
```

Out[32]:
```
<Axes: title={'center': 'Cumulative Return 2018-2026'}, xlabel='Date'
>
```



In [33]:
```
log_returns.rolling(window=90).std().plot(title='Rolling 90-Day Volati
```

Out[33]:
```
<Axes: title={'center': 'Rolling 90-Day Volatility'}, xlabel='Date'>
```



Correlation Matrix

In [35]:
```
log_returns.corr()
```

Out[35]:

Ticker	XLB	XLC	XLE	XLF	XLI	XLK	XLP
Ticker							
XLB	1.000000	0.646519	0.675911	0.706546	0.824636	0.614886	0.583809
XLC	0.646519	1.000000	0.412002	0.663857	0.666752	0.776353	0.522465
XLE	0.675911	0.412002	1.000000	0.582137	0.638567	0.446655	0.448478
XLF	0.706546	0.663857	0.582137	1.000000	0.787813	0.624240	0.572597
XLI	0.824636	0.666752	0.638567	0.787813	1.000000	0.736374	0.623750
XLK	0.614886	0.776353	0.446655	0.624240	0.736374	1.000000	0.441295
XLP	0.583809	0.522465	0.448478	0.572597	0.623750	0.441295	1.000000
XLRE	0.647176	0.576871	0.424604	0.624653	0.659498	0.527758	0.671172
XLU	0.521762	0.410007	0.475293	0.494032	0.552083	0.431792	0.602085
XLV	0.630207	0.610142	0.473182	0.621854	0.704082	0.626628	0.614156
XLY	0.721803	0.804859	0.508681	0.739198	0.802626	0.735980	0.588977

Task 4

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Written Assesment

Presented here is a dynamic and interactive data analysis document. The user can build a portfolio of tickers to perfrom analysis and run the coding cells producing analysis. It all works sequentially, so one can perform another analysis by just going back to the top and choosing a different date range and set of tickers.

Comparing 2-3 stocks, intially I chose Apple, Lockheed Martin, and TORM (A shipping tanker company). These different tickers exhibit different return and volatility characteristics. You can see that Lockheed Martin exhibits the lowest volatility, as a large stable company in the defense sector, this makes sense. Compared to Apple in the technology sector, we can see Apple's higher-cumulative return matches their higher volatility.

From a macroeconomic perspective you can see the impact of large exogenous events on all of these stocks. Most notably 2000 era Dot-Com crash in Apple.

Volatility spikes and negative returns in 2007,2008 due to the Great Financial Crisis. You can also note 2020 COVID-era crash and subsequent rally, followed by 2022 interest-rate hiking adding volatility. More recently you can see 2025 "Liberation Day" volatility spikes from President Trump's aggressive tarrif policy announcements. As an tanker company that deals in oil-markets, you can see Torm's stock rally during 2022 Russian oil embargo, and sinking of oil vessels in Red Sea and 2026 war on Iran. TRMD's exposure to these regional crises has been limited, while their company has benefited from the higher-price point demanded in international markets for shipping tankers.

In [ ]: